

HAIYANG, China

WMO: 548630

Lat: 36.7750N

Lon: 121.1782E

Elev: 40

StdP: 100.85

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.5	-6.9	-18.4	0.7	-1.0	-16.6	0.9	-1.8	10.6	-0.7	9.7	-0.3	2.8	290	0.436

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 5.7	(c) 31.2	(d) 24.2	(e) 29.9	(f) 24.0	(g) 28.9	(h) 23.7	(i) 27.0	(j) 29.0	(k) 26.4	(l) 28.3	(m) 25.7	(n) 27.6	(o) 3.8	(p) 180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.5	22.1	28.4	25.8	21.2	27.8	25.2	20.4	27.1	85.8	29.1	82.9	28.4	80.0	27.6	29.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.6	8.3	7.2	-10.5	33.8	1.7	1.7	-11.7	35.0	-12.7	36.0	-13.6	37.0	-14.8	38.2
			-11.5	27.7	1.5	0.9	-12.6	28.3	-13.5	28.8	-14.3	29.3	-15.5	29.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	13.1	-0.8	1.2	5.9	12.0	17.8	21.9	25.4	25.8	21.8	15.8	8.3	1.7
	DBStd	9.74	3.17	3.24	3.36	3.00	2.84	2.17	1.95	1.92	2.32	3.18	3.99	3.39
	HDD10.0	1067	336	247	132	14	0	0	0	0	0	2	78	258
	HDD18.3	2604	594	481	384	192	44	2	0	0	3	90	300	516
	CDD10.0	2206	0	0	6	73	242	356	478	488	354	181	28	0
	CDD18.3	702	0	0	0	1	28	108	219	230	106	9	0	0
	CDH23.3	4721	0	0	0	7	156	444	1632	1962	495	26	0	0
	CDH26.7	1030	0	0	0	2	29	91	364	480	63	2	0	0
Wind	WSAvg	3.2	3.2	3.2	3.5	3.6	3.4	3.0	3.0	3.0	2.8	2.9	3.2	3.1
Precipitation	PrecAvg	778	10	8	19	45	60	83	206	196	84	35	23	9
	PrecMax	1661	52	35	71	126	198	248	498	611	343	103	140	32
	PrecMin	391	0	0	0	1	6	6	49	33	2	1	1	0
	PrecStd	250	13	9	17	34	50	66	117	125	72	30	29	10
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.0	11.7	18.4	23.2	29.7	31.6	33.1	32.8	29.7	25.7	19.6	12.5
		MCWB	5.3	5.8	9.8	13.5	17.2	20.1	25.2	25.2	21.6	17.3	14.1	9.7
	2%	DB	7.0	9.4	15.2	20.6	26.9	29.1	31.2	31.1	28.3	23.7	17.7	10.3
		MCWB	3.9	4.7	8.7	12.6	16.3	19.8	25.3	24.8	20.9	16.5	12.8	6.9
	5%	DB	5.5	7.7	13.0	18.6	24.9	27.3	29.8	30.0	27.1	22.2	16.2	8.6
		MCWB	2.4	4.0	7.0	11.3	15.9	19.6	24.9	24.7	20.6	16.2	12.3	5.5
	10%	DB	4.1	6.3	11.1	16.9	23.1	25.8	28.6	29.1	26.1	20.8	14.6	6.9
		MCWB	1.3	3.2	6.1	10.7	15.4	19.6	24.6	24.7	20.1	15.5	11.0	4.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.4	8.2	11.8	16.4	20.2	24.2	27.9	27.7	25.4	20.3	16.5	10.8
		MCDB	7.6	9.5	15.6	20.6	25.4	27.1	30.6	29.9	27.2	22.8	18.1	12.0
	2%	WB	4.5	6.2	9.7	14.6	18.9	22.9	27.0	27.2	23.8	18.9	14.7	7.8
		MCDB	6.1	7.9	13.0	17.9	23.3	24.9	29.1	29.1	25.9	21.6	16.3	9.7
	5%	WB	3.0	4.9	8.5	13.4	17.9	22.1	26.3	26.7	22.7	17.9	13.0	6.0
		MCDB	4.9	6.9	11.3	16.2	21.5	24.3	28.3	28.5	25.2	20.7	15.3	8.1
	10%	WB	1.6	3.6	7.3	12.4	17.0	21.3	25.6	26.1	21.8	16.8	11.4	4.3
		MCDB	3.9	5.8	10.0	15.1	20.4	23.7	27.5	27.9	24.5	19.7	14.3	6.6
Mean Daily Temperature Range	5% DB	MDBR	7.2	7.6	8.0	7.9	7.9	6.3	5.0	5.7	7.4	8.2	7.8	7.1
		MCDBR	8.8	9.1	10.3	10.6	11.4	9.6	7.0	6.7	8.4	9.2	8.9	9.0
		MCWBR	6.2	5.9	6.0	5.5	4.8	4.0	3.2	3.0	3.8	4.7	5.8	6.4
	5% WB	MCDBR	8.1	7.7	9.1	8.2	8.5	6.2	5.1	4.5	6.2	7.5	7.5	8.1
		MCWBR	6.5	5.9	6.0	5.1	4.7	3.3	2.8	2.5	3.7	4.9	6.2	6.7
Clear-Sky Solar Irradiance	taub		0.482	0.538	0.591	0.612	0.636	0.730	0.709	0.634	0.557	0.580	0.549	0.490
	taud		1.817	1.732	1.655	1.664	1.657	1.532	1.631	1.769	1.878	1.733	1.747	1.821
	Ebn at Noon		649	668	680	699	692	628	637	675	703	622	578	605
	Edh at Noon		161	198	233	244	249	282	254	217	184	195	172	151
All-Sky Solar Radiation	RadAvg		2.39	3.17	4.49	5.23	5.68	5.26	4.65	4.67	4.32	3.48	2.51	2.10
	RadStd		0.21	0.31	0.33	0.28	0.41	0.43	0.49	0.51	0.34	0.30	0.25	0.14

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	+0.92	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+81	+51

Nomenclature: See separate page